

3.5 inch LCD TFT
(2.4 英寸真彩色液晶显示屏)
320RGBx480 Resolution and 65K color
(240x320 分辨率, 6 万 5 千种色)

TABLE OF Contents 目录

1. General Descripti 基本描述.....	3
2. Mechanical Specification 机械规格.....	3
3. Mechanical Dimension 机械尺寸图.....	4
4. Electrical Maximum Ratings 电气极限.....	5
5. Brightness characteristic&Power dissipation 亮度特性&功耗.....	5
6. Module Function Description 显示屏脚位定义.....	6
7. Response time&Contrast ratio 响应时间和对比度.....	10
8. Viewing Angle 视角宽度.....	11
9. Reliability Trial 可靠性实验.....	12
10. Inspection Standards 检验标准.....	12
11. Package Method 包装方法.....	14

1.General Description 基本描述

MODEL NO 产品型号	ZJY350S11CTG21
Display Mode 显示模式	Transmissive 全透
Display Format 显示格式	Graphic 320RGB*480 Dot-matrix 320xRGBx480 图形点阵
Input Data 显示屏接口类型	4 线串口
Viewing Direction 视角方向	12 o'clock 12 点钟
Drive 显示屏驱动芯片	ST7796U

2. Mechanical Specification 机械规格

Item	Specifications	Unit
Dimensional outline 显示屏外围尺寸	57.26(W)*96.52(H)*5.48(T) (带触摸)	mm
Resolution 分辨率	320RGB*480	dots
LCD Active area 显示尺寸	48.96(W)*73.44 (H)	mm
Pixel size 像素尺寸	0.153(W)*0.153(H)	mm



商城地址: <https://zjy-display.taobao.com> TEL: 18601955397

Figure 1: Mechanical drawing of the LCD module. The drawing includes three views: a top view showing the display area with dimensions like 159.3, 57.26, and 55.26; a side view showing the module's profile with a height of 15.00; and a bottom view showing the pin connections. A table of pin descriptions is provided, listing pins 1 through 11 and their functions: GND, VCC, CLK, MOS, RES, DC, CS, BLK, SCL, SDA, and INT. The drawing also includes various mechanical tolerances and material specifications.

4. Electrical Maximum Ratings 电气极限

Item 项目	Symbol 符号	Min 最小值	Max 最大值	Unit 单位	Note 备注
Supply voltage (VCC) 工作电压(VDD)	V	2.8	3.3	V	-
Operating temperature 工 作温度范围	T _{OPR}	-20	70	℃	-
Storage temperature 存储温度范围	T _{STR}	-30	80	℃	-

※NOTE: VDDI 和 VDD 可以直接连一起, 共用一组 (2.8V~3.3V) 电压供电。

5. Brightness characteristic&Power dissipation 亮度特性&功耗

Item 项目	Symbol 符号	Min 最小值	Typical 典型值	Max 最大值	Unit
LED module Forward voltage LED 背光源正向电压	V _{LED}	2.9	3.1	3.3	V
LED module current LED 背光源电流	I _{LED}	-	60	-	mA
LCD Surface Luminance 显示屏表面亮度	L _S	250	300	-	Cd/m ²
LCM Surface brightness uniform LED 背光源均匀度	L _D	80	-	-	%
LCD power dissipation 显示屏总功耗	P _{LCD}	-	0.22	-	W

※NOTE: $P_{LCD} = VDD * (I_{LED} + I_{LCD})$

6.1 Module Function Description 显示屏脚位定义

引脚序号	引脚名称	作用描述	备注
1	GND	接地脚	-
2	VCC	显示屏电源供电脚 2.8-3.3V	-
3	CLK	SPI 串口的时钟信号	-
4	MOSI	SPI 串口的数据输入脚	-
5	RESET	显示屏复位脚, 低电平复位	
6	DC	4 线 SPI 串口的显示数据/寄存器指令	
7	CS	显示屏驱动芯片选脚, 低电平使能	
8	BLK	背光控制管脚, 低电平关闭背光, 默认为高电平	
9	SCL	电容触摸屏 IIC_SCL 信号	
10	SDA	电容触摸屏 IIC_SDA 信号	-
11	INT	电容触摸屏中断信号	-

※备注:1.给背光源供电时, 需要在背光源正极或负极接限流电阻, 使背光源的总电流限制在 90mA 以内, 避免长时间使用时因电流过大发热, 造成显示屏永久损坏。背光源的限流很重要, 规格书里反复提醒。

2.不管是并口模式还是 SPI 串口模式下, 显示数据传输顺序都是高位在前。

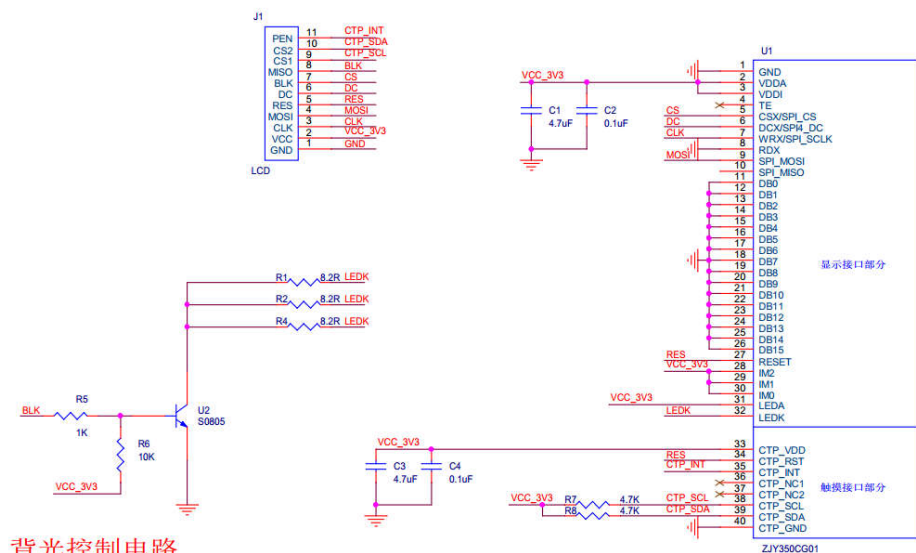
6.2 Capacitive touch screen interface definition.电容触摸屏接口定义

引脚序号	引脚名称	作用描述	备注
9	SCL	电容触摸屏 IIC_SCL 信号	-
10	SDA	电容触摸屏 IIC_SDA 信号	-
11	INT_3.3V	电容触摸屏中断信号	-

※备注:1.触摸控制芯片型号位 GT911

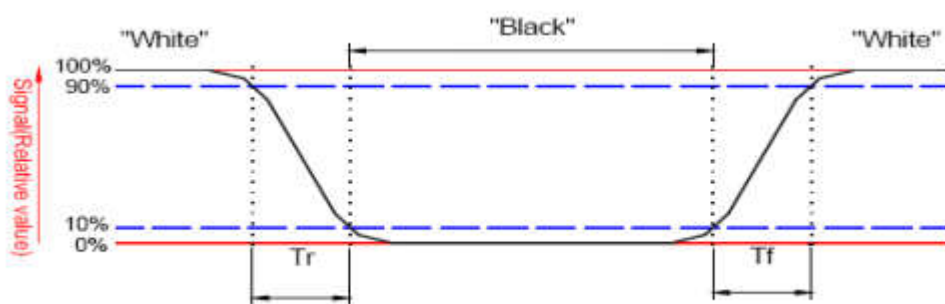
2.触摸最大支持 5 点触摸

附图 6-1: SPI 接口参考应用电路



7. Response time & Contrast ratio 响应时间与对比度

Item 项目	Symbol 符号	Condition 条件	Remark			Unit 单位
			Min. 最小值	Typ. 典型值	Max. 最大值	
Response time 响应时间	Tr+Tf	$\theta = 0^\circ$	-	25	40	ms
Contrast ratio 对比度	CR	$\theta = 0^\circ$	350	500	-	-



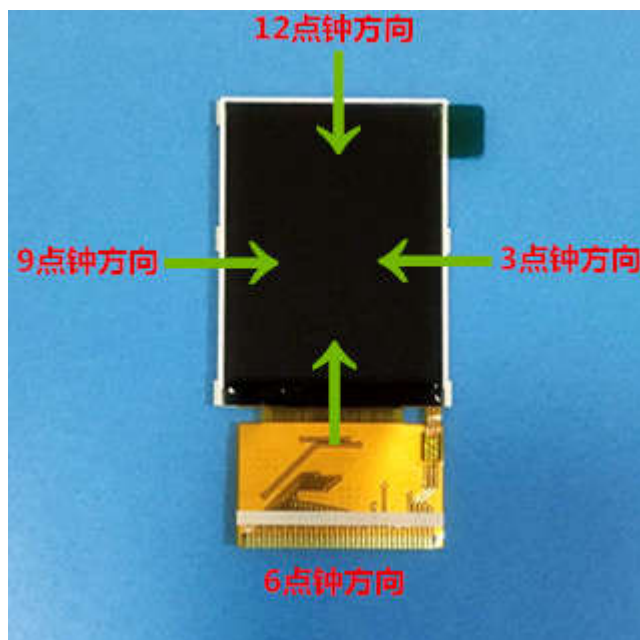
响应时间图示

$$\text{Contrast ratio (CR)} = \frac{\text{Brightness on the "white" state}}{\text{Brightness on the "black" state}}$$

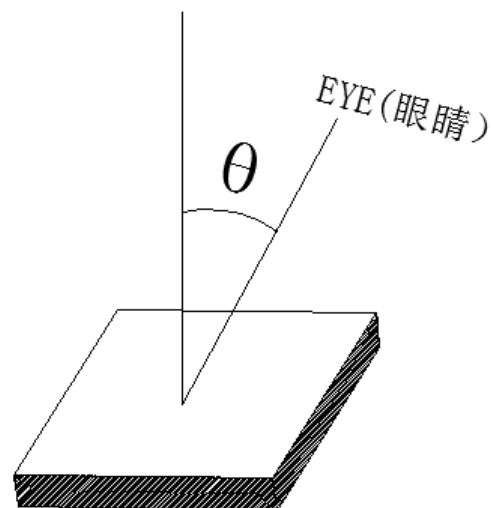
对比度计算公式

8.Viewing Angle 视角宽度

Item 项目	Symbol 符号	Condition 条件	Remark			Unit 单位
			Min. 最小值	Typ. 典型值	Max. 最大值	
Viewing angle 视角宽度	Top 12 点钟方向	CR≥10 对比度大于等于 10	40	50	-	Deg. 度
	Bottom 6 点方向	CR≥10 对比度大于等于 10	55	65	-	
	Left 9 点钟方向	CR≥10 对比度大于等于 10	55	65	-	
	Right 3 点钟方向	CR≥10 对比度大于等于 10	55	65	-	



垂直于屏表面



NOTE:3 点, 6 点, 9 点, 12 点方向视角的大小指的是垂直于屏表面的线眼睛视线之间的夹角(θ)。

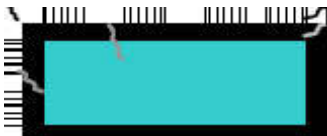
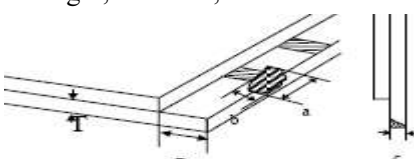
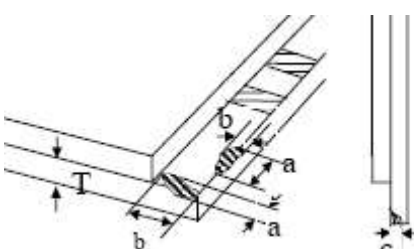
9. Reliability Trial 可靠性实验

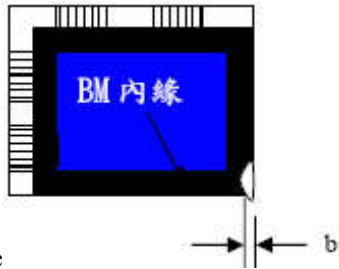
NO. 序号	ITEM 实验项目	CONDITION 实验环境	CRITERION 实验规范
1	High Temperature Non-Operating Test 高温存储实验	80℃*120Hrs	No Defect Of Operational Function In Room Temperature Are Allowable 室温运行功能无缺陷
2	Low Temperature Non-Operating Test 低温存储实验	-30℃*120Hrs	
3	High Temperature/Humidity Non Operating Test 高温高湿实验	60℃*90%RH*120Hrs	
4	High Temperature Operating Test 高温工作实验	70℃*72Hrs	
5	Low Temperature Operating Test 低温工作实验	-20℃*72Hrs	
6	Thermal Shock Test	-20 ℃ (30Min) ↔ 70 ℃	

	热冲实验	(30Min) *10CYCLES	
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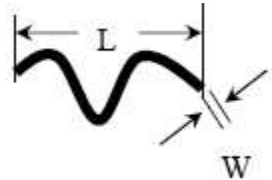
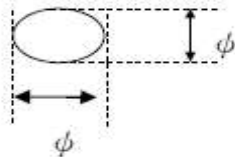
10. Inspection standards 检验标准

10.1 Glass defect

NO	Defect item	Criteria	Remark
1	Dimension Unconformity (Major defect)	By Engineering Drawing	
2	Cracks (Major defect)	1. Linear cracks panel 【Reject】 2. Nonlinear crack contrast by limited sample	
3	Glass extrude the conductive area (minor defect)	a: disregards and no influence assemblage. 1) $b \leq 1/3$ Pin width (non bonding area) 【Accept】 2) bonding area ≤ 0.5 mm 【Accept】	A: Length, b: Width
4	Pin-side ,conductive area damaged (minor defect)	(a c: disregards) $b \leq 1/3$ of effective length for bonding electrode 【Accept】	a: length, b: Width, c: Thickness 
5	Pin-side, non-conductive area damaged (minor defect)	1) Damage area don't touch the ITO (Including contraposition mark, except scribing mark) 【Accept】 2) $C < T$ $b \leq BM/3$ of width 【Accept】 3) $c = T$ b not touch the seal glue 【Accept】 4) a disregards	a: Length, b: Width c: Thickness 
6	Non-pin-side damage (minor defect)	$c < T$ 1) b exceeds $1/3 B_m$ 【Reject】 $c = T$ b not touch the seal glue 【Reject】	c: Thickness b: width of

			
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10.2LCD appearance defect(View area)

NO	Defect item	Criteria		Remark
1	Fiber、glass cratch、polarizer scratch/folded (minor defect)	Specification	Allowable	note1:L: Length, W: Width note2: disregard if out of AA 
		$W \leq 0.03\text{mm}$	disregard	
		$0.03\text{mm} < W \leq 0.05\text{mm};$ $L \leq 3.0\text{mm}$	2	
		$0.05\text{mm} < W \leq 0.1\text{mm};$ $L \leq 3.0\text{mm}$	1	
		$W > 0.1\text{mm}; L > 3.0\text{mm}$	0	
2	Polarizer bubble、 concave and convex (minor defect)	$\phi \leq 0.2\text{mm}$	disregard	note1: $\phi = (L+W)/2$, L:Length, W :Width note2:disregard if out of AA
		$0.2\text{mm} < \phi \leq 0.3\text{mm}$	2	
		$0.3\text{mm} < \phi \leq 0.5\text{mm}$	1	
		$0.5\text{mm} < \phi$	0	
3	Black dots、dirty dots、 impurities、eye winker (minor defect)	$\phi \leq 0.15\text{mm}$	disregard	note2:disregard if out of AA 
		$0.15\text{mm} < \phi \leq 0.25\text{mm}$	2	
		$0.25\text{mm} < \phi \leq 0.3\text{mm}$	1	
		$0.3\text{mm} < \phi$	0	
4	Polarizer prick (minor defect)	$\phi \leq 0.1\text{mm}$	disregard	note1: $\phi = (L+W)/2$, L=Length, W=Width note2:the distance between two dots>5mm
		$0.1\text{mm} < \phi \leq 0.25\text{mm}$	3	
		$\phi > 0.25\text{mm}$	0	

11.Package Method 包装方法

显示屏出货包装示意图：

